

BIT-Triazine Acrylate Resins as High-Performance Sustainable Coatings for Marine Antifouling Applications

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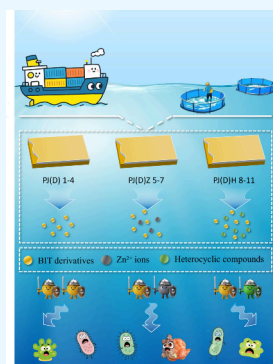
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ABSTRACT: This study presents the rational design and performance evaluation of eco-friendly BIT-triazine acrylic resins for marine antifouling applications. Two monomers were synthesized using cyanuric chloride as a core scaffold, incorporating the effective biocide 1,2-benzisothiazolin-3-one (BIT) and a polymerizable methallyl alcohol group. A total of 22 copolymers were prepared via copolymerization with acrylic esters and laboratory-synthesized functional monomers. Their chemical structures were confirmed by FT-IR and ^1H NMR spectroscopy. Hydrolytic degradation tests revealed that polymers containing zinc monomers showed the highest contact angle change ($\Delta\theta > 20^\circ$), while those with heterocyclic monomers exhibited the fastest surface erosion rates ($0.0138 \text{ mg/cm}^2/\text{day}$), indicating tunable degradation behavior. Biological assays demonstrated excellent antifouling performance: polymers with pyridin-2-yl acrylate achieved algae inhibition rates of 94.47–99.93% and antibacterial rates of 98.61–99.69% against various marine strains. Importantly, these polymers exhibited low or negligible toxicity toward mussels and barnacle larvae. In a 90 day marine field test, all resins showed a strong antifouling efficacy, particularly those containing zinc or heterocyclic monomers. This work establishes a modular design strategy for multifunctional antifouling resins that are adaptable to diverse marine environments, including ships, aquaculture systems, and submerged infrastructure.

KEYWORDS: BIT-triazine, high performance, sustainable, antifouling, coating



1. INTRODUCTION

Marine biofouling refers to the phenomenon in which various organisms in the ocean get attached and grow on the surface of objects.^{1,2} These organisms include bacteria, algae, barnacles, shellfish, and other marine organisms. Biofouling can occur on man-made structures such as ships, docks, nets, and pipelines as well as on natural marine organisms and surfaces. This phenomenon increases the resistance and fuel consumption of ships, accelerates corrosion on the surface of artificial facilities structures, blocks the cooling circulation system, and destroys the marine ecological balance through the reproduction of invasive species.^{3–5} Traditional antifouling strategies rely on toxic coatings containing tributyltin (TBT) or cuprous oxide, which have been banned or gradually restricted due to their durable environmental toxicity, thus requiring the development of environmentally friendly alternatives.^{6–8}

Currently existing environmentally friendly antifouling agents are mainly divided into natural products and synthetic antifouling agents. Natural product antifouling agents have the advantages of being nontoxic, efficient, rapid degradation, and environmentally harmless. However, their disadvantages are the low content of active ingredients, high extraction cost, and difficulty in large-scale production.^{6–11} Insufficient supply is a major problem in the commercialization of natural product antifoulants. In order to make up for the shortcomings of natural product antifoulants and avoid the application obstacles caused by their limitations, artificial synthetic antifoulants have

emerged and have been widely studied and applied.¹² Artificial synthetic antifoulants are a type of natural product antifoulant that contain natural product structures or their analogs through chemical means. Its advantage is that its active ingredients are clear, the structure can be accurately regulated to enhance antifouling efficiency, and it has a mature production process, low cost, long antifouling cycle, etc. In particular, isothiazolinone compounds exhibit broad-spectrum antibacterial activity and environmental compatibility and have good practicality.^{13–19} These compounds interfere with microbial DNA replication through the N–S bond and then rapidly degrade seawater to prevent the bioaccumulation. Nevertheless, limitations in aqueous solubility and synthetic complexity hinder their industrial application.

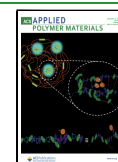
To address these challenges, 1,3,5-triazine derivatives have attracted our attention because of their good stability and antimicrobial,^{20–23} antiviral,^{24,25} and antitumor applications,^{26–28} along with antituberculosis efficacy.^{29,30} Cyanuric chloride (2,4,6-trichloro-1,3,5-triazine) is an important 1,3,5-triazine derivative,^{31,32} and its derivative Irgarol 1051 has been

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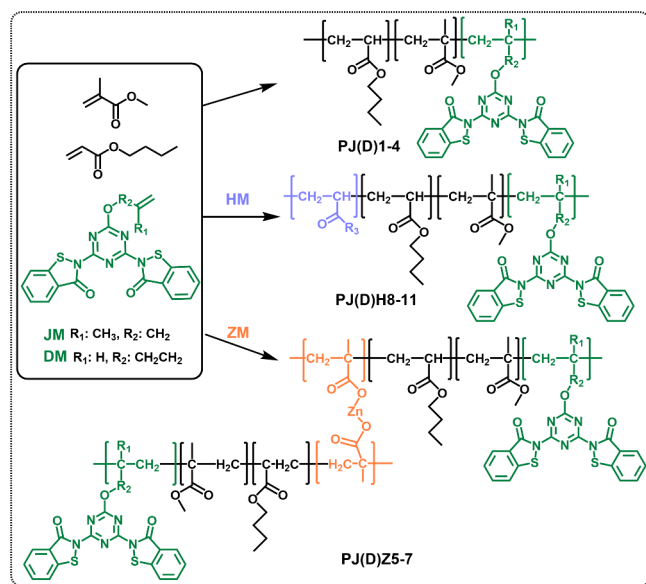
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proven to be an effective marine antifouling agent.^{33–35} Based on the excellent antifouling effect of isothiazolinones and Irgarol 1051, we synthesized two antifouling monomers (JM and DM) by combining their structural characteristics and using cyanuric chloride as the reaction scaffold and isothiazolinones as the antifouling group.

These monomers were achieved through functionalization of the cyanuric chloride with bactericidal 1,2-benzisothiazoline-3-one (BIT) moieties to enhance antifouling capabilities. Subsequently, these monomers underwent copolymerization with either two acrylate co-monomers, a bifunctional zinc monomer (ZM), or heterocyclic monomers (HM) to fabricate antifouling resins (PJ and PD series) (Scheme 1). The resultant

Scheme 1. Synthesis Routes of PJ and PD Polymers



polymers underwent rigorous physicochemical characterization to determine critical parameters: ¹H NMR, FT-IR, the number-average molecular weight (M_n), the glass transition temperatures (T_g), the thermal gravimetric analysis (TGA), hydrolysis contact angle change, and mass loss rate. The biological evaluation system encompassed bacteriostatic efficacy quantification, algal growth suppression analysis, antiadhesion efficacy assessment against two representative fouling organisms (barnacle larvae and mussel), and systematic validation of composite antifouling performance through standardized marine field tests.

2. EXPERIMENTAL SECTION

2.1. Chemicals. BIT, cyanuric chloride, methallyl alcohol, 3-butenol, sodium bicarbonate, triethylamine, xylene, *n*-butanol, methyl methacrylate (MMA), butyl acrylate (BA), and methacrylic acid (MAA) were purchased from Aladdin. All of the chemicals were used directly without further purification. The five compounds, i.e., benzo[d]oxazol-2-yl acrylate (HM1), pyridin-2-yl acrylate (HM2), *N*-(6-chlorobenzo[d]thiazol-2-yl) acrylamide (HM3), *N*-(6-methylbenzo[d]thiazol-2-yl) acrylamide (HM4), and bifunctional zinc (ZM) were prepared in the laboratory. The specific experimental details, materials, and methods were presented in the [Supporting Information](#). The synthetic route involving compounds is shown in [Supporting Information](#) Figure S1. ¹H NMR spectrum data and IR spectrum data of related compounds were shown in [Figures S4–S12](#).

2.2. Preparation of JM and DM Monomers. In a 100 mL three-neck flask, cyanuric chloride (9.25 g, 0.05 mol), sodium bicarbonate

(5.88 g, 0.07 mol), and methallyl alcohol (30 mL) were combined and stirred at 0–5 °C for 1 h, followed by continued stirring at room temperature for 3 h. Upon completion, the reaction mixture was extracted with ethyl acetate (60 mL), dried over anhydrous MgSO₄, and filtered to afford a colorless solution of 2,4-dichloro-6-(methallyloxy)-1,3,5-triazine. This solution was then placed in a 250 mL reaction flask containing BIT (15.1 g, 0.1 mol). After the mixture was stirred at room temperature for 30 min, triethylamine was added dropwise to neutralize the liberated HCl. The reaction mixture was stirred for an additional 3 h, and completion was confirmed by TLC. The resulting mixture was extracted with ethyl acetate, washed with water, and concentrated under reduced pressure to yield a yellow powdery solid. The yield is 92.90%, and the melting point is 115.6–117.2 °C. **JM**, ¹H NMR (400 MHz, CDCl₃): δ 7.90 (d, 2H, PhH), 7.81 (d, 2H, PhH), 7.60 (t, 2H, PhH), 7.45 (t, 2H, PhH), 4.85 (d, 2H, =CH₂), 4.67 (s, 2H, CH₂), 1.69 (s, 3H, CH₃). IR (KBr), ν (cm⁻¹): 3072, 2974, 2921, 1715 (C=O), 1596, 1563, 1502, 1448 (CH₂), 1387 (CH₃), 904, 738.

The synthetic route refers to the above steps. The yield is 88.65%, and the melting point is 118.7–120.1 °C. **DM**, ¹H NMR (400 MHz, CDCl₃): δ 7.78 (d, 2H, PhH), 7.73 (d, 2H, PhH), 7.48 (t, 2H, PhH), 7.34 (t, 2H, PhH), 5.62 (m, H, =CH=), 4.97 (d, 2H, =CH₂), 4.26 (t, 2H, CH₂), 2.35 (q, 2H, CH₂). IR (KBr), ν (cm⁻¹): 3070, 2976, 2927, 1728 (C=O), 1589, 1557, 1495, 1462 (CH₂), 1399 (CH₃), 987, 919, 740.

2.3. Preparation of PJ and PD Copolymers. In a 250 mL three-necked flask, a mixed solvent of xylene and *n*-butanol was added, and the mixture was heated at about 85 °C in an oil bath. Separately, a monomer feed solution was prepared by dissolving JM monomer (5.0 g), methyl methacrylate (MMA, 47.5 g), butyl acrylate (BA, 47.5 g), and AIBN (3.21 wt % relative to total monomers) in the same solvent mixture (total volume, 60 mL). After complete dissolution, the solution was loaded into a constant-pressure dropping funnel and added dropwise to the reactor over 2.5 h under controlled feeding conditions. Following the addition, the reaction mixture was maintained at 85 °C for an additional 3 h. Then, a second portion of AIBN (1.38 wt % relative to total monomers) was added to ensure complete polymerization. Reaction progress was monitored using TLC to confirm the consumption of the JM monomer. Upon completion, a light yellow polymer solution was obtained and labeled as PJ1. Additional polymers (PJ2–PJH11 and PD1–PDH11) were synthesized using the same procedure, varying only in monomer feed compositions.

2.4. Material Characterizations. The nuclear magnetic resonance hydrogen spectra (¹H NMR) of all compounds and polymers were tested using a BRUKER av-400 nuclear magnetic resonance spectrometer. The Fourier transform infrared spectroscopy (FT-IR) of all compounds and polymers was tested using a TENSOR27 infrared spectrometer. The molecular weight and distribution of the polymer were measured using a Waters 1525 (Waters Co., USA) gel permeation chromatograph. The glass transition temperature (T_g) of the polymer was obtained with a Q100 SDT differential scanning calorimeter from TA Inc., USA. A Q600 SDT thermogravimetric analyzer from TA Inc., USA was used to analyze the thermogravimetric changes of the polymer. The contact angle change of ultrapure water on the surface of polymer after hydrolysis was tested by an SL200 KB interfacial tension meter from Shanghai SUOLUN Co. The mass change of polymer after hydrolysis was weighed by an AL204 electronic balance from Shanghai Mettler Toledo Co. The adhesion of the polymer coating was tested using a KT-500Z fully automatic adhesion tester from Shenzhen Ketan Electronic Technology Co., Ltd. The hardness of the polymer coating was tested using the QHQ-A pencil hardness meter from Zhejiang Airuipu Instrument Co., Ltd.

2.5. Algae Growth Inhibition Experiment. Diatoms are the main algae composition in marine biological pollution communities. *C. curvisetus* is selected as the research object for algae inhibition performance evaluation. This study further selected *Chlorella* and *I. galbana*, a typical species of the *Chlorophyta* and the representative species of the *Chrysophyta*, as model organisms and systematically evaluated the broad-spectrum inhibitory effect of polymer coating on the attachment behavior of multiple algae species. The polymers were coated on glass plates (76.2 mm × 25.4 mm), the control group was

pure acrylic resin PJ(D)0, and the blank group was uncoated glass slides. They were naturally dried at room temperature and placed in a 230 mL plastic cup. Under the illumination of LED tubes (12 h light/12 h dark cycle, 5000–6000 lx intensity), 180 mL of diluted experimental algae solution was placed in each plastic cup, and the control group as well as blank group samples were tested in the same way. The absorbances of the sample and the control group were measured and recorded every 24 h.

2.6. Bacterial Growth Inhibition Test. Marine bacterial communities are predominantly Gram-negative. To systematically evaluate antimicrobial efficacy, we selected *V. corallilyticus* and *V. parahaemolyticus* as model organisms, supplemented with Gram-positive *S. aureus* to assess the coating's broad-spectrum antimicrobial potential. The antibacterial effect of the resin was analyzed by the absorbance method. First, 1 mL of the resin was evenly applied to a cell culture plate (6 wells) and dried at room temperature until no solvent was left. Before testing, the cell culture plate coated with resin needs to be sterilized by ultraviolet light (UV) for 2 h. Then, 2 mL of culture medium and 15 μ L of well-cultured bacterial solution were added and cultured in a constant temperature incubator at 37 °C. Every 2 h, 20 μ L of bacterial solution was aspirated into a 96-well cell culture plate, and the absorbance changes were measured at 600 nm using a microplate reader for 2, 4, 6, 8, 10, and 12 h, while the changes in the bacterial solution in the cell culture plate were observed and recorded. The growth inhibition rate of each sample on the bacterial solution was calculated using the following formula:

$$T = \frac{A_0 - A_i}{A_0 - 0.047} \times 100\% \quad (2-1)$$

where T is the growth inhibition rate of bacteria, A_0 is the maximum concentration of blank bacterial solution, A_i is the concentration of sample bacterial solution, and 0.047 is the average absorbance value of the holes in the blank culture plate.

2.7. Large Fouling Organism Growth Inhibition Test. Barnacles and mussels were collected in Haikou Bay, Haikou City. The following process was used for the inhibition of barnacle larvae attachment experiment: a pipet was used to take 1 mL of each polymer sample that was then applied evenly on a cell culture dish and dried at room temperature. A 15 mL portion of fresh seawater was pipetted into the culture dish, and 15 active barnacle larvae were put into it. Culturing was done at room temperature, and the number of barnacle larvae attached after 12 and 24 h was observed and recorded. For the mussel growth inhibition experiment, the following steps were used: A 1 mL aliquot of resin was pipetted, spread evenly on a glass culture dish, and allowed to dry at room temperature. A 15 mL aliquot of filtered seawater was added to each culture dish, and 3 mussels were placed in each culture dish. A culture dish without any substance was used as a blank control. The number of mussel byssus threads were observed after 24 h. After 24 h of the experiment, the mussels were removed and put back into new glass culture dishes. A 15 mL portion of fresh sterilized seawater was added to each new culture dish. After 12 h, the number of byssus threads of the mussels in the culture dish was recorded.

2.8. Marine Field Test. The experimental site was located in Haikou Bay, Haikou City, Hainan Province (20°01'N, 110°19'E), with seawater salinity ranging from 29.6 to 31.8‰. The experiment was carried out according to the national standard GB 5370-2007 (method for testing antifouling panels in shallow submergence). The soaking time is from July to September, which lasts for 3 months. The polymer sample was brushed on the surface of the steel plate twice, dried naturally, and then immersed in seawater at a depth of 0.5–1.0 m. Natural product antifoulant 5-octyl-2-furanone (C1),^{36,37} commercial antifouling agent cuprous oxide (C2), and acrylic resin PJ(D)0 were used as controls, respectively. After every one month, the steel plates were taken out of the sea, cleaned with seawater, and photographed. They were then immediately returned to seawater for further testing. The antifouling activity was evaluated according to ASTM D6990-05 (2011).³⁸

3. RESULTS AND DISCUSSION

3.1. Material Characterizations. The number-average molecular weights (M_n) of polymers PJ and PD are about 10000–15000 (Table 1 and Table S1). The obvious increase in

Table 1. Characterization of PJ Polymers

Sample	Weight ratio of monomer ^a	<i>M</i> _n (g/mol)	PDI	<i>T</i> _g (°C)
	MMA/BA/JM/ZM/HM			
PJ1	47.5/47.5/5/0/0	9992	2.00	59.06
PJ2	45/45/10/0/0	10052	2.01	58.52
PJ3	42.5/42.5/15/0/0	10240	1.96	53.57
PJ4	40/40/20/0/0	9999	2.02	60.71
PJZ5	37.5/37.5/20/5/0	10799	2.05	46.29
PJZ6	35/35/20/10/0	14397	1.73	63.33
PJZ7	32.5/32.5/20/15/0	15387	1.68	71.85
PJH8	32.5/32.5/20/0/15(HM1)	10765	1.94	71.77
PJH9	32.5/32.5/20/0/15(HM2)	10285	1.98	50.89
PJH10	32.5/32.5/20/0/15(HM3)	11294	1.90	61.46
PJH11	32.5/32.5/20/0/15(HM4)	10836	1.94	65.60
PJ0	50/50/0/0/0/0	9858	2.16	58.03

^aPercent basis (%).

molecular weight indicated that polymerization has occurred. The polymer dispersion index (PDI) was between 1.68 and 2.16, indicating that the molecular weight distribution was relatively uniform.³⁹ Each copolymer exhibits a single glass transition temperature T_g (DSC curves of the polymer PJH9 were shown in Figure S2), indicating that all polymers are amorphous. If T_g is too low or too high, the coating performance will not meet the routine requirements. The T_g values of the above 22 copolymers are within the ideal range, and they can easily form films.

The structures of the synthesized polymers and monomers were confirmed using ¹H NMR and FT-IR spectroscopy. Figure 1a displays the ¹H NMR spectrum of polymer PJ4. Characteristic aromatic proton signals appeared at δ = 7.20–8.25 (position d), corresponding to the benzene ring protons from the JM moiety. Peaks at δ = 4.65 and 4.67 (position c) were attributed to the CH₂ group adjacent to the oxygen in the JM monomer. Signals at δ = 3.75 and 4.15 (position a) and δ = 3.45 and 3.65 (position b) correspond to CH₂–O in the BA monomer and CH₃–O in the MMA monomer, respectively. Additional signals at δ = 1.50–1.60 and δ = 1.25–1.40 (positions a₃ and a₂) were assigned to CH₂ groups in MMA, while δ = 1.40–1.45 (position e) corresponding to CH₃ groups directly attached to quaternary carbon atoms in MMA and JM. A distinct signal at δ = 0.78–0.96 (position a₁) was attributed to terminal CH₃ groups from BA. The key structural difference between polymer PJ4 and compound JM4 lies in the presence of a terminal carbon–carbon double bond in the monomer. Figure 1b shows the ¹H NMR spectrum of compound JM, where a characteristic peak at δ = 4.82–4.88 (position f) corresponds to vinylic CH₂ protons, which is absent in PJ4, confirming successful polymerization and consumption of the double bond. Similarly, panels c and d of Figure 1 show the ¹H NMR spectra of polymer PD4 and DM monomer, respectively. In Figure 1d, vinylic proton signals at δ = 5.57 and 5.67 and δ = 4.92 and 5.00 are clearly observed, confirming the presence of the double bond in the DM monomer. The absence of these signals in the spectrum of PD4 (Figure 1c) indicates the complete incorporation of DM into the polymer backbone.

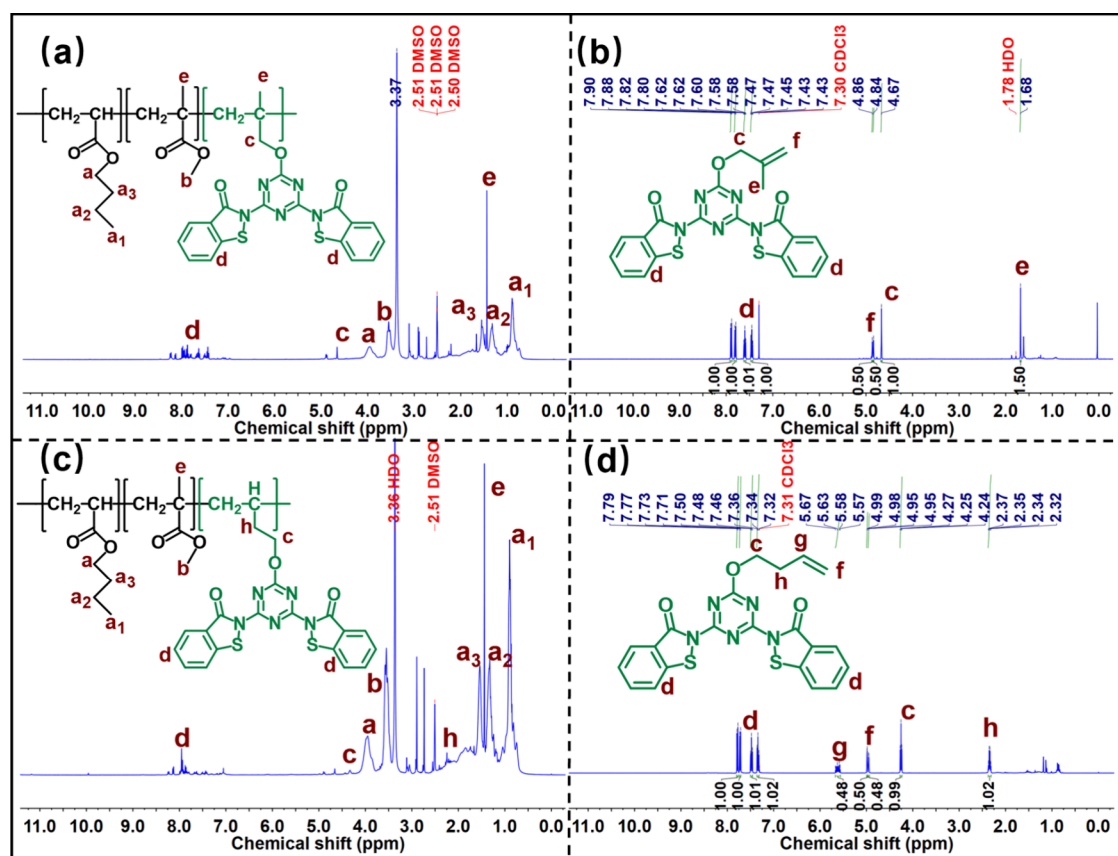


Figure 1. ^1H NMR spectra of polymer PJ4 (a), compound JM (b), polymer PD4 (c), and compound DM (d).

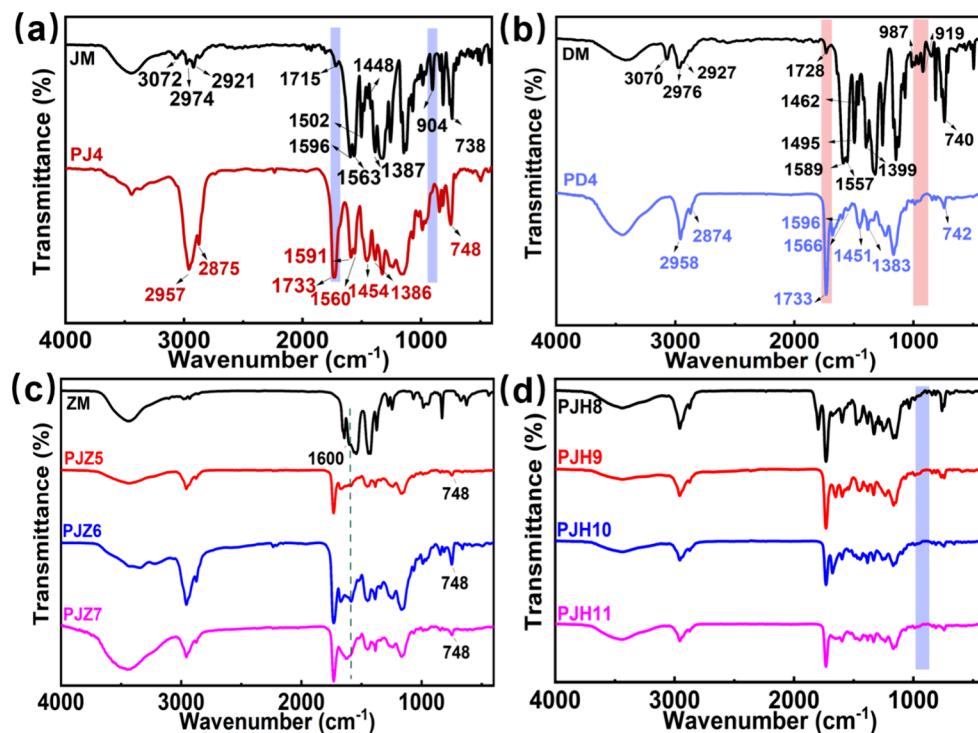


Figure 2. FT-IR spectra of polymer PJ4 and compound JM (a), polymer PJ4 and compound JM (b), polymers PJZ5–7 and compound ZM (c), and polymers PJH8–11 (d).

Figure 2a shows the FT-TR spectrum of polymer PJ4 and compound JM. We first analyze the infrared data of compound JM. 738 cm^{-1} can be attributed to the ortho-disubstituted

structure of BIT in the JM monomer itself. The peak at 904 cm^{-1} can be attributed to the characteristic absorption peak of olefin structure $\text{R}_2\text{C}=\text{CH}_2$ in the JM monomer. The 1387 and 1448

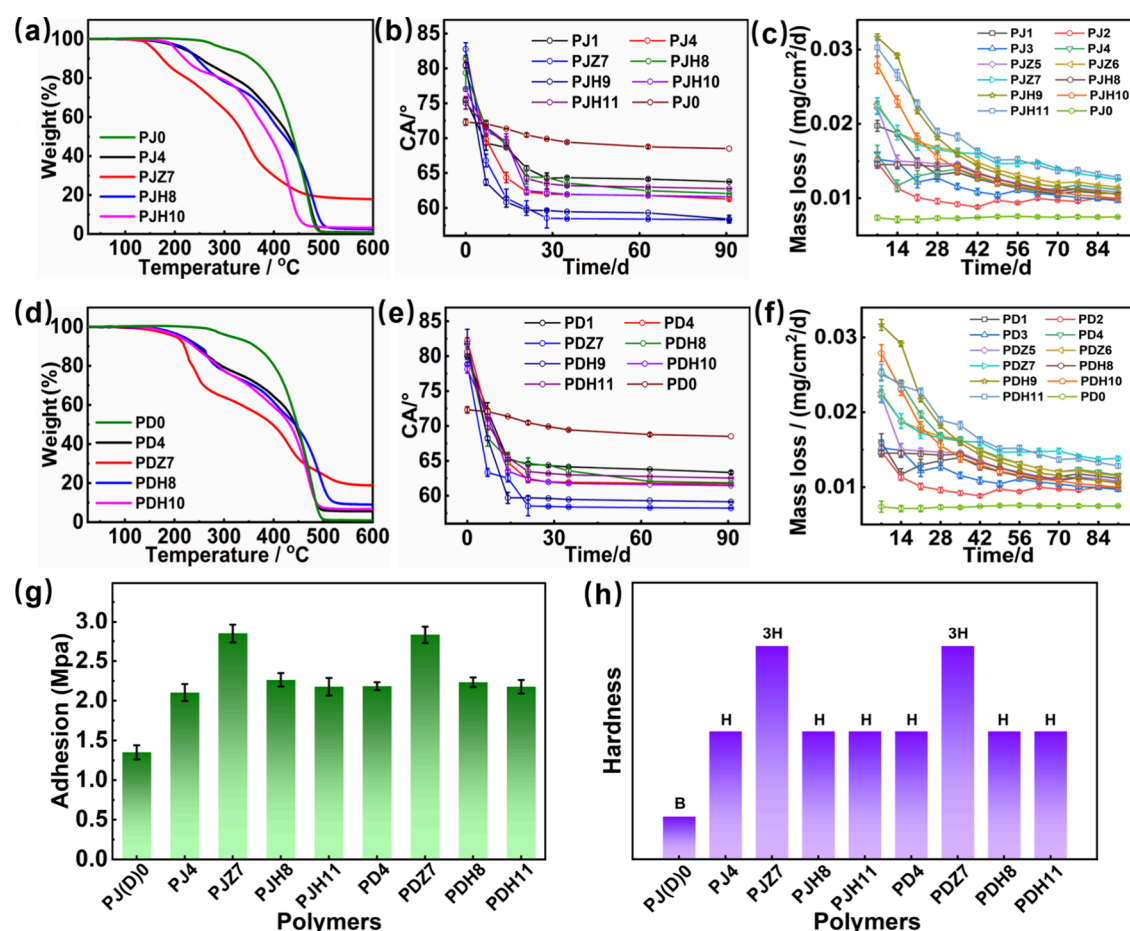


Figure 3. (a) TGA curve of polymer PJ. (b) Contact angle change of polymer PJ before and after hydrolysis. (c) Hydrolysis mass loss rate of polymer PJ. (d) TGA curve of polymer PD. (e) Contact angle change of polymer PD before and after hydrolysis. (f) Hydrolysis mass loss rate of polymer PD. (g) Adhesion of polymer coatings. (h) Hardness of polymer coating.

cm^{-1} peaks can be attributed to the characteristic absorption peaks of methyl and methylene in JM monomer. The 1502 and 1563 cm^{-1} peaks can be attributed to the benzene ring structure of BIT in the JM monomer. The 1715 cm^{-1} peak can be attributed to the keto carbonyl structure of BIT in the JM monomer itself. Peaks 2974 and 2921 cm^{-1} can be attributed to the characteristic absorption peaks of methyl and methylene in JM monomer. Peak 3072 cm^{-1} can be attributed to the characteristic absorption peak of the olefin structure $\text{R}_2\text{C}=\text{CH}_2$ and the benzene ring structure in the JM monomer. After polymerization, polymer PJ4 showed a group of new peaks at 1733 cm^{-1} , which can be attributed to the characteristic absorption peaks of the ester groups in MMA and BA monomers in the resin. Peak 748 cm^{-1} can be attributed to the ortho-disubstituted structure of BIT in the JM monomer itself. There are no characteristic absorption peaks of the olefin structures $\text{R}-\text{CH}=\text{CH}_2$ and $\text{R}_2\text{C}=\text{CH}_2$ in the range of 900–1000 cm^{-1} . The results showed that there is no olefin structure in the polymer, indicating that the polymer is successfully copolymerized.

Figure 2b shows the FT-IR spectrum of polymer PD4 and compound DM. We first analyze the infrared data of compound DM. 740 cm^{-1} can be attributed to the ortho-disubstituted structure of the benzene ring in the DM monomer. Values 987 and 919 cm^{-1} can be attributed to the characteristic absorption peaks of the olefin structure $\text{R}-\text{CH}=\text{CH}_2$ in the DM monomer. After polymerization, the resin PD4 has a new group of peaks at 1733 cm^{-1} , which can be attributed to the characteristic

absorption peaks of the ester groups in MMA and BA in the resin. Peak 742 cm^{-1} can be attributed to the ortho-disubstituted structure of the benzene ring in the DM monomer. There are no characteristic absorption peaks of the olefin structures $\text{R}-\text{CH}=\text{CH}_2$ and $\text{R}_2\text{C}=\text{CH}_2$ in the range of 900–1000 cm^{-1} . The results show that there is no olefin structure in the polymer, indicating that the polymer is successfully copolymerized.

Figure 2c shows the infrared spectra of polymers PJZ5–7 and compound ZM. The 1600 cm^{-1} reading is attributed to the characteristic absorption peak of the O–Zn–O structure in compound ZM;⁴⁰ 748 cm^{-1} can be attributed to the JM monomer itself having an ortho-disubstituted structure of BIT. This indicated that the polymer has been successfully prepared. Figure 2d shows the infrared spectrum of polymers PJH8–11. There is no characteristic absorption peak of olefin structure $\text{R}-\text{CH}=\text{CH}_2$ in the range 900–1000 cm^{-1} (blue area). The results showed that there is no olefin structure in the polymer, indicating that the polymer is successfully copolymerized. The infrared spectrum data of polymers PDZ5–7 and polymers PDH8–11 are similar to it and will not be described here.

3.2. Physical and Chemical Properties of Coatings.

Thermogravimetric analysis (TGA) of the resin was conducted over 25–600 °C under nitrogen atmosphere, with a heating rate of 10 °C/min. Panels a and d of Figure 3 are the thermal stability curves of polymers PJ and PD, respectively. The thermogravimetric changes of polymers can be divided into two processes. The first process ranges from about 150 to 250 °C, and the

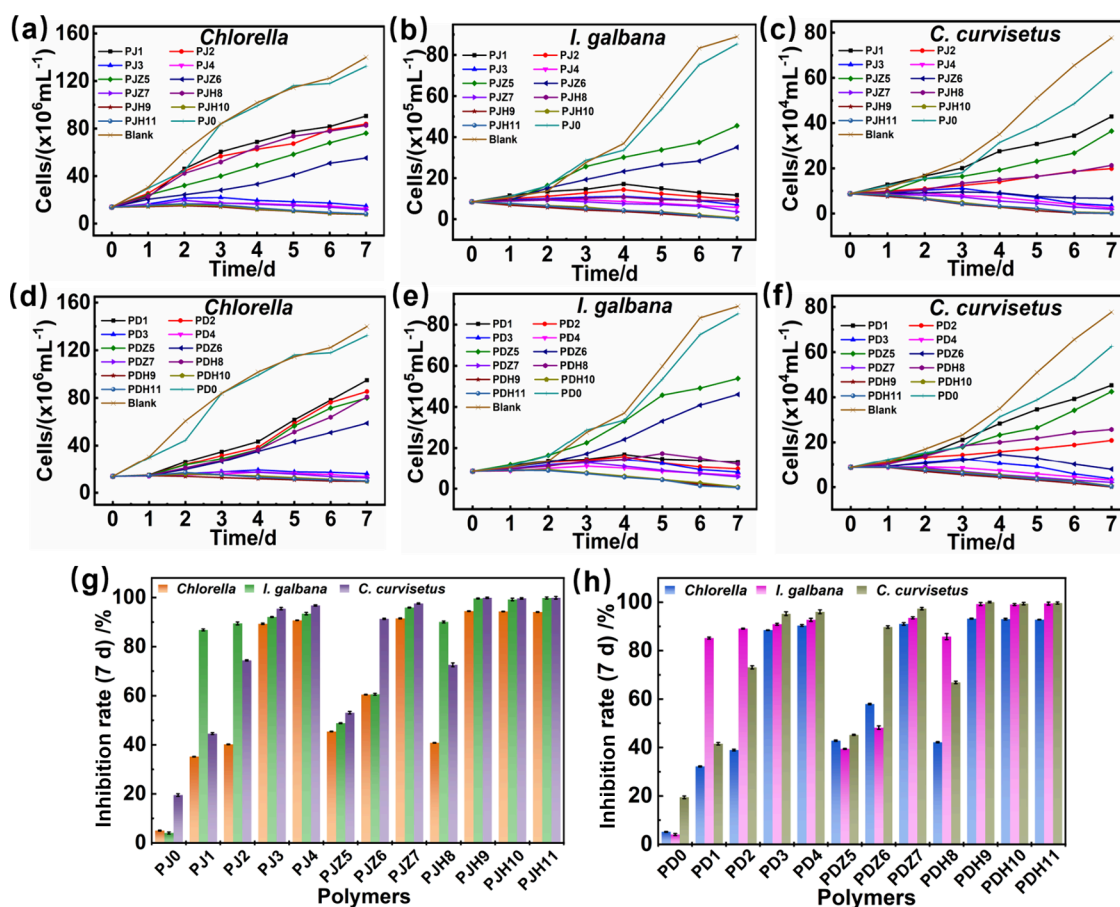


Figure 4. Growth inhibition of *Chlorella* (a), *I. galbana* (b), and *C. curvisetus* (c) by polymer PJ. Growth inhibition of *Chlorella* (d), *I. galbana* (e), and *C. curvisetus* (f) by polymer PD. Growth inhibition rate of three algae species by polymer PJ (g) and polymer PD (h).

weight loss rate during this process is about 20%. The possible reason for this is that the unreacted small molecules are decomposed by heat during the reaction. The second process range is about 250–500 °C. In this process, the polymer weight loss rate is relatively large, close to 80%, because the copolymer decomposes under heat. It begins to decompose rapidly at 300 °C, and after 500 °C, the mass loss of the copolymer is close to 100%, proving that the polymer is almost completely vaporized. The polymers PJZ7 and PDZ7 are exceptions. Their weight loss rate in the second process is about 60%. Most of the polymers are decomposed and about 20% remains. This is because zinc carboxylate exists in the polymer components, and zinc cannot be decomposed. All polymers comply with the typical rules of thermal decomposition of acrylic resins.^{41,42}

Panels b and e of Figure 3 show the contact angle changes of polymers PJ and PD before and after hydrolysis, respectively. Before hydrolysis, the contact angle of the polymer prepared by adding the JM or DM monomer is significantly larger than that of pure acrylate PJ(D)0, indicating that the added monomer improves the hydrophobicity of the polymer. When placed in sterilized seawater, the contact angle of polymer PJ(D)0 changes very little, because after hydrolysis there is almost no effect on the chemical structure of the copolymer surface. After the introduction of various monomers, the contact angle of the polymer decreased with an increase of the hydrolysis time. After 21 days, the change in the contact angle tended to be stable. This is because after the JM monomer, zinc monomer ZM or heterocyclic monomer HM on the polymer side chain is hydrolyzed, the surface becomes a hydrophilic surface, and the

final angle is approximately between 58 and 65°. Among them, the contact angle changes of PJZ7 and PDZ7 before and after hydrolysis are the largest, indicating that the introduction of bifunctional zinc monomer ZM accelerates the hydrolysis process of the polymer. The contact angle changes of PJH9 and PDH9 before and after hydrolysis are also quite obvious, indicating that the introduction of the heterocyclic monomer HM also accelerates the hydrolysis process of the polymer.

Panels c and f of Figure 3 show the hydrolysis mass loss rates of polymers PJ and PD. As can be seen from Figure 3c, the mass loss rate of polymer PJ0 is almost unchanged with an average rate of 0.005 mg/cm²/day. The main reason for the mass change is the hydrolysis of the ester group. Compared with polymer PJ0, the rate of mass loss increases with the addition of more JM monomers, indicating the increases of hydrolysis ability of the sample with the amount of added monomers. The reason for the mass change of polymers PJ1–4 is mainly due to the hydrolysis of the JM monomer. For polymers PJZ5–7, the weight loss of the samples is mainly due to the self-polishing rate of zinc monomer ZM and the hydrolysis of JM monomer. For polymers PJH8–11, the weight loss of the samples is mainly due to the hydrolysis of the heterocyclic compound HM and the hydrolysis of JM. The results showed that the introduction of different monomers will affect the hydrolysis rate of polymer PJ. In particular, polymer PJH11 with the addition of heterocyclic monomer HM4 has the fastest hydrolysis rate, with an average rate of 0.0138 mg/cm²/day. As can be seen from Figure 3f, the hydrolysis rate of polymer PDH9 is the fastest, with an average rate of 0.0138 mg/cm²/day.⁴⁴

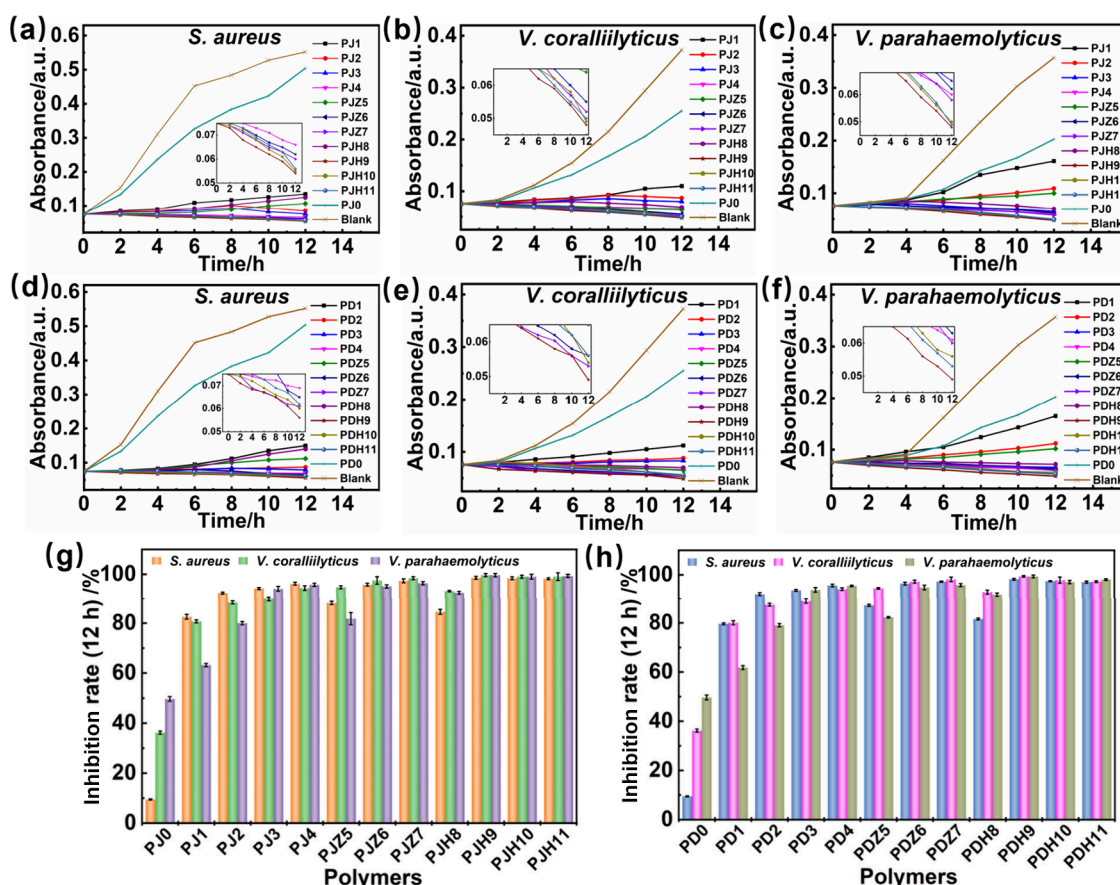


Figure 5. Growth inhibition of *S. aureus* (a), *V. coralliilyticus* (b), and *V. parahaemolyticus* (c) by polymers PJ. Growth inhibition of *S. aureus* (d), *V. coralliilyticus* (e), and *V. parahaemolyticus* (f) by polymers PD. Inhibition rate of growth of three bacteria by polymers PJ (g) and polymers PD (h).

Panels g and h of Figure 3 respectively present the adhesion and hardness test results of the prepared marine antifouling coating. Both are the core indicators for evaluating the mechanical properties of the coating and directly determine their long-term service capabilities in a marine environment. The adhesion of the coating was determined by the pull-off test according to the international standard (ISO 4624:2023), and the results showed that the adhesion between the coating and substrate was stable at 2–3 MPa. It can resist water flow erosion and vibration impact, avoiding early peeling. At the same time, the adhesion range of 2–3 MPa achieves a balance control of binding strength and coating toughness, which not only prevents brittle cracking of the coating caused by excessive highs but also avoids binding failure caused by excessive lows, so that it can adapt to complex working conditions from nearshore low flow velocity areas to medium and high flow velocity sea areas.

The hardness was determined by a pencil hardness method according to international standards (ISO 15184:2012), and the result was H-3H. It can meet the scratch resistance requirements of the light attachment areas of algae, can withstand high-strength mechanical effects such as barnacle removal, and can cover the common load range of marine biological physical removal (corresponding to H-4H). This range is consistent with the industry's mainstream antifouling coating hardness standard (H-4H), confirming that it can meet the surface damage resistance needs of ship hulls, marine platforms, and other equipment during service.

In conclusion, the combined properties of adhesion and hardness work synergistically to create a robust protective

system. Adhesion ensures the structural integrity of the coating, while hardness enhances resistance to localized surface damage. Together, they provide a stable platform for antifouling functionalities, including controlled antifouling agent release and the prevention of biological adhesion. This integrated approach meets the diverse needs of marine engineering applications across multiple environments.

3.3. Evaluation of the Inhibitory Activity of Polymers against Algae Growth. We selected *Chlorella*, *I. galbana*, and *C. curvisetus* as research objects to evaluate the algae inhibitory properties of polymers.^{45,46} The absorbance value of the algae was tested with an ultraviolet–visible spectrophotometer every 24 h. The algae solution concentration was calculated based on the algae absorbance value–concentration working curve in Figure S3, and the inhibition curve of algae solution concentration (C)–time (d) was drawn, while the algae inhibition rate was calculated using the algae solution concentration on the seventh day. Figure 4 shows the inhibition curves and inhibition rates of polymers PJ and PD against *Chlorella*, *I. galbana*, and *C. curvisetus*.

Panels a and d of Figure 4 show the inhibitory ability of polymers PJ and PD on the growth of *Chlorella*, respectively. Compared with the control group PJ0 and blank group, polymers PJ and PD both had certain algae inhibitory activity. As the JM monomer content increased from 5 to 20%, the inhibitory activity was significantly enhanced. Compared with the blank group, polymers PJ3 and PJ4 had a strong inhibitory ability on the growth of *Chlorella*, and the concentration remained basically unchanged after slow growth, with the

highest inhibition rate of 90.87%. The inhibitory effects of polymers PJZ5 and PJZ6 on *Chlorella* were significantly worse than those of polymer PJZ7. The reason is that the trace amounts of zinc ions can promote plant growth, but when zinc ions reach up to a certain concentration, they can inhibit their growth.^{47,48} Polymers PJH9–11 and PDH9–11 containing heterocyclic monomers basically inhibited the growth of *Chlorella* in an early stage. It showed significant inhibitory activity against *Chlorella*, with the highest inhibition rate of 94.47%.

Panels b and e of Figure 4 show the inhibitory ability of polymers PJ and PD on the growth of *I. galbana*, respectively. When the algae solution was cultured for 7 days, all polymers PJ and PD were observed to significantly inhibit the growth of *I. galbana*. The average inhibition rates of polymers PJZ7 and PDZ7 were 95.97 and 93.53%. Polymers PJH9–11 and PDH9–11 containing heterocyclic monomers showed significant inhibitory activity against the growth of *I. galbana*, with the highest inhibition rate of 99.83%.

Panels c and f of Figure 4 show the inhibitory activity of polymers PJ and PD on the growth of *C. curvisetus*, respectively. On the first day of culture, there was no significant difference in the concentrations of *C. curvisetus* in each test group. During the subsequent days of cultivation, the inhibitory activity significantly changed. The results showed that polymers PJ and PD exhibited good inhibitory effects on the growth of *C. curvisetus*. The inhibition rate of polymer PJZ5 on *C. curvisetus* was less than 60%, while the average inhibition rates of polymers PJZ6 and PJZ7 were 91.34 and 97.59% respectively. The resins polymerized with heterocyclic monomers showed significant inhibitory properties. In particular, polymers PJH9–11 and PDH9–11 showed excellent inhibitory activity against the growth of *C. curvisetus* with the highest inhibition rate of 99.93%.

Overall, the introduction of zinc monomers and heterocyclic monomers can significantly improve the inhibitory ability of the polymer against algae. The inhibitory ability of polymers PJ and PD on the three algae was slightly different, while the inhibitory effect on *I. galbana* and *C. curvisetus* was significantly better than that on *Chlorella*. The reason is that *Chlorella* itself grows faster and the concentration of algae liquid is high, so the inhibitory effect is slightly worse than that of the other two algae. The antialgal mechanism of the polymeric system originates from the hydrolysis-mediated release of BIT derivatives, Zn^{2+} ions, and heterocyclic compounds. The inhibitory mechanism of BIT derivative is achieved by destroying the photosynthetic system of algae. The inhibited algae strain can recover when its growth is not completely inhibited. This inhibition has little to no effect on subsequent algae regeneration. It is worth noting that even if the algae growth is completely inhibited, the algae will not die.⁴⁹ The released Zn^{2+} ions or heterocyclic compounds played a synergistically inhibit algae.

3.4. Evaluation of the Inhibitory Activity of Polymers against the Growth of Marine Bacteria. To assess the antibacterial performance of the synthesized polymers, both Gram-positive bacteria (*S. aureus*) and Gram-negative bacteria (*V. corallilyticus* and *V. parahaemolyticus*) were selected as test organisms. As shown in Figure 5, synthesized polymers exhibited measurable inhibitory effects against bacterial growth. In contrast, the blank control (pure bacterial solution) showed rapid growth with a corresponding increase in absorbance over time. The presence of copolymers consistently suppressed bacterial proliferation throughout the experimental period, indicating an effective antibacterial activity.

Panels a–c of Figure 5 display the bacterial growth inhibition curves for the PJ-series polymers. Antibacterial efficacy improved with increasing JM monomer content, with PJ4 demonstrating the highest inhibition rates of 96.24, 94.48, and 95.81% against *S. aureus*, *V. corallilyticus*, and *V. parahaemolyticus*, respectively. Incorporation of bifunctional zinc monomers in PJZ5–PJZ7 (at 5, 10, and 15% loading) further enhanced antibacterial activity, with PJZ6 and PJZ7 outperforming PJ4. Similarly, polymers modified with heterocyclic monomers (PJH9–PJH11) exhibited strong antibacterial effects, particularly PJH9, which achieved inhibition rates of 98.61, 99.69, and 99.68% against the three tested bacterial strains. These results underscore the synergistic contribution of JM, zinc, and heterocyclic monomers in enhancing the antibacterial efficacy.

Panels d–f of Figure 5 show the inhibition curves of polymer PD on the growth of three bacterial solutions. As the DM monomer content increases, the antibacterial effect becomes more obvious, and the best antibacterial effect is polymer PD4, with an average inhibition rate of 95.64, 94.17, and 95.48%. The antibacterial ability of polymers PDZ6 and PDZ7 prepared by adding 10 and 15% bifunctional zinc was very significant, both better than PD4. After adding heterocyclic monomers (PDH9, PDH10, and PDH11), polymers PDH9, PDH10, and PDH11 significantly inhibited the activity of bacterial broth, especially polymer PDH9, with inhibition rates of 98.22, 99.39, and 99.35% on the three bacterial broths, respectively.

To summarize, the inhibitory effects of polymers PJ and PD on *V. corallilyticus* and *V. parahaemolyticus* are better than those on *S. aureus*. The reason is that *S. aureus* grows and reproduces relatively fast, while *Vibrio* grows and reproduces slowly, so the inhibitory effect is slightly worse than that on the other two bacteria. The results showed that the polymer had a stronger inhibitory effect on Gram-negative bacteria than on Gram-positive bacteria.^{15,50} The above polymer is hydrolyzed to generate BIT derivatives, Zn^{2+} ions, or heterocyclic compounds. The thiol groups on BIT derivatives can interact with the bases of DNA in organisms to form hydrogen bonds, which prevents the organisms from replicating, inactivates proteins, and affects the normal growth of bacteria, ultimately leading to cell apoptosis.¹⁶ Since the bacterial surface is negatively charged, the positively charged metal zinc ions are adsorbed on the bacterial surface, thereby destroying its electrolyte balance and causing the bacteria to die due to damage to the cell wall.⁵¹ Heterocyclic compounds themselves have strong antibacterial properties, and their mechanism of action is unique and not exclusive. They have the characteristics of broad spectrum, high efficiency, and safety.⁵² As core pharmacodynamic groups, the benzothiazole ring and pyridine ring mainly involve interference with key physiological processes of bacteria and have multitarget characteristics. Both types of compounds have certain hydrophobicity.⁵³ Benzothiazole rings contain sulfur atoms, and pyridine rings are electron-deficient systems. They can be inserted into bacterial cell membranes (especially phospholipid bilayers) through hydrophobic or electrostatic effects. This insertion will disrupt the ordered structure of the membrane, increase membrane permeability, lead to leakage of important ions (such as K^+ and Ca^{2+}) and metabolites in the cell, destroy transmembrane potential, affect energy metabolism and material transportation, and ultimately lead to cell lysis or death. Compounds containing benzothiazole or pyridine structures can be embedded in the double helix structure of bacterial DNA or bind to specific bases of DNA/RNA, interfering with DNA replication, transcription, or RNA translation processes. It is

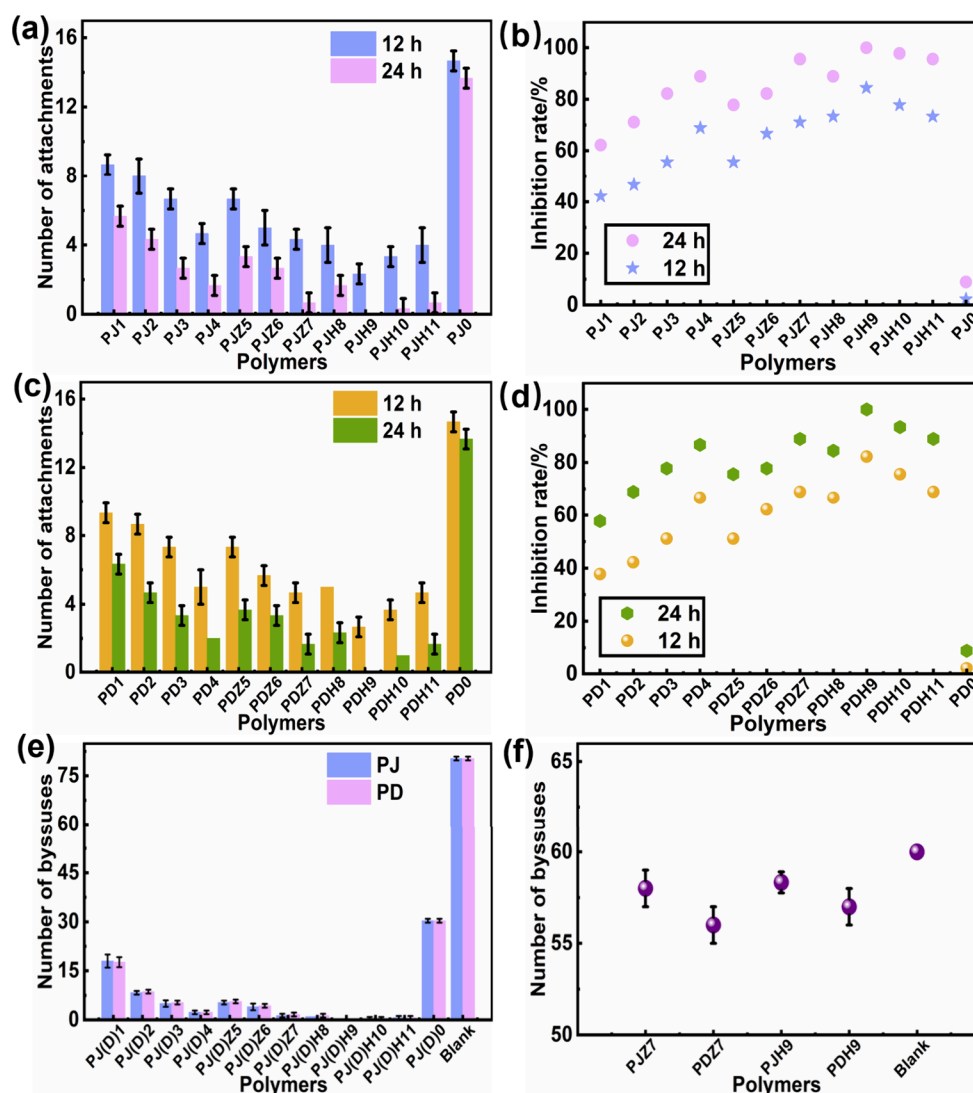


Figure 6. Effects (a) and inhibition rates (b) of polymers PJ on the attachment of barnacle larvae. Effects (c) and inhibition rates (d) of polymers PD on the attachment of barnacle larvae. (e) Number of byssus threads secreted by mussels in the test solution for 24 h (e) and in fresh seawater for 12 h (f).

noteworthy that chlorine substitution significantly affects the activity, with chlorine substitution being more effective than methyl substitution.⁵⁴

3.5. Evaluation of Inhibitory Activity against Large Fouling Organisms. In order to investigate the inhibitory effect of polymer coatings on the attachment of large fouling organisms, we used the number of attached barnacle larvae and the number of byssus threads secreted by mussels to evaluate the inhibitory activity.

Panels a and b of Figure 6 show the inhibition trend of polymer PJ on the attachment of barnacle larvae. Compared with the control group PJ0, the introduction of JM monomer into the copolymer molecule showed higher inhibitory activity. When the mass proportion of JM monomers reached 20% (PJ4), the inhibition rate of barnacle larvae attachment increased to 66.67% at 12 h and 86.67% at 24 h. When 15% zinc monomer was added to the polymer PJZ7, the inhibition rate of barnacle larvae attachment increased to 71.11% at 12 h and up to 95.56% at 24 h. The inhibition rates of barnacle larvae attachment of polymers PJH9 and PJH10 prepared by grafting heterocyclic monomers were increased to 84.44 and 77.78% at 12 h, and reached 100.0 and 97.78% at 24 h, respectively. Panels c and d of

Figure 6 show the inhibitory trend of polymers PD on the attachment of barnacle larvae. Compared with polymers PJ, the inhibitory activity trend of polymer PD on barnacle larvae was slightly poorer. The mortality of polymers PD4, PDZ7, and PDH9 on barnacle larvae increased to 86.67, 88.89, and 100.0% at 24 h, respectively.

The above results showed that the polymer has a significant inhibitory effect on the attachment of barnacle larvae. In summary, the introduction of bifunctional zinc monomers and heterocyclic compounds improves the inhibitory activity of the polymer on the attachment of barnacle larvae.

Figure 6e shows the effects of polymers PJ and PD on the number of byssus threads secreted by mussels for 24 h. The polymers PJ(D)Z7 with the introduction of zinc monomers and PJ(D)H8–11 with the introduction of heterocyclic monomers have a more significant effect on the number of byssus threads secreted by mussels. Especially for polymer PJ(D)H9, mussels did not secrete byssus, indicating that the polymer had a significant inhibitory effect on the number of byssus secreted by mussels.¹⁷ In order to test whether the polymers have toxic effects on mussels, we chose to remove the mussels from polymers PJ(D)Z7 and PJ(D)H9, put them back into fresh

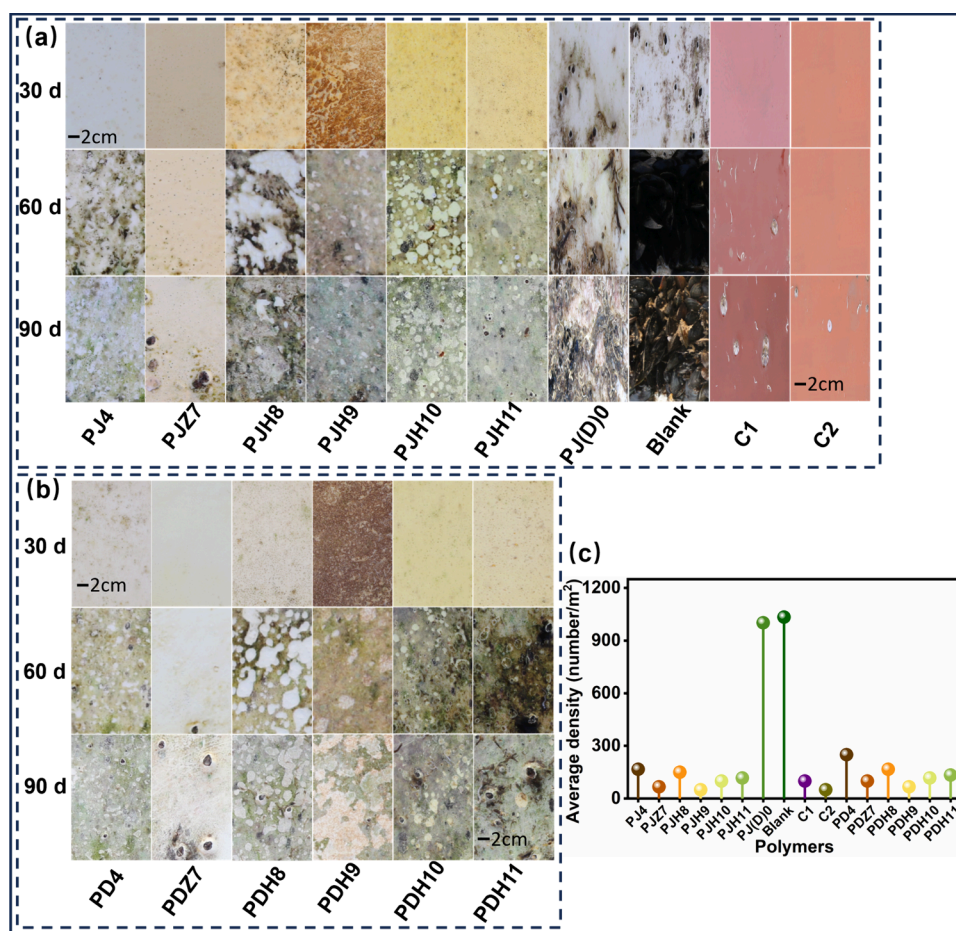


Figure 7. (a) Changes of polymers PJ on the marine field test after 90 days. (b) Changes of polymers PD on the marine field test after 90 days. (c) Average barnacle density on the experimental plates after immersion of polymers PJ and PD for 90 days.

seawater, and observe the number of byssus threads they secreted.

Figure 6f shows the number of byssus threads that were secreted by mussels after 12 h. It can be seen from the figure that the activity of mussel byssus threads secreted by fresh seawater was not affected, indicating that the synthetic polymer only inhibited the secretion of mussel byssus threads and had no effect on their activity. It showed that the synthesized polymer has very low toxicity to organisms and is expected to become an environmentally friendly antifouling resin.

3.6. Marine Field Test Experiment Evaluation. To assess the practical antifouling capabilities of the synthesized polymers, marine field tests were conducted over a 90 day period. Polymer coatings from the PJ and PD series were applied to primed steel plates and submerged in seawater. The tested samples included polymers containing 20% JM (PJ4, PJZ7, and PJH8–PJH11) or DM monomers (PD4, PDZ7, and PDH8–PDH11). Control samples consisted of PJ(D)0, without any antifouling monomers, C1, a natural antifouling formulation containing 5-octylfuran-2(SH)-one, and C2, a commercial antifouling coating with cuprous oxide.

Fouling observations over time are presented in Figure 7a,b. By day 30, barnacle attachment was evident on the blank and PJ(D)0 samples, while C1 and C2 remained clean. All of the test polymer coatings showed no visible biofouling at this stage. After 60 days, the blank steel plates were heavily fouled with mussels, barnacles, and oysters. PJ(D)0 samples exhibited some barnacles and coiled tubeworms but notably resisted mussel

attachment, suggesting a mild inherent antifouling effect of the acrylic resin. A few tubeworms appeared on C1, while C2 showed no fouling. The test polymer coatings showed minor green algae growth, with the minimal presence of larger fouling organisms. At the end of 90 days, the blank plates were completely covered with barnacles and mussels. PJ(D)0 surfaces showed increased colonization by barnacles and tubeworms. C1 had a slight increase in fouling, and C2 showed only trace amounts. In contrast, test polymers maintained strong resistance to biofouling with PJZ7 and PDZ7, with zinc-containing monomers having minimal barnacle attachment and no algal growth. In the same way, PJH11 and PDH8–PDH10, featuring heterocyclic monomers, outperformed PJ4 and PD4 in resisting biofouling. These findings demonstrate that incorporating zinc and heterocyclic functional groups significantly improves the antifouling performance. The chemical structure of the polymer side chains plays a key role in preventing the attachment of marine organisms, making these coatings suitable for various long-term marine applications.

The average density of barnacle attachment in the polymer test group was 50–250 per square meter, and the average density of barnacle attachment in the blank group was 1034 per square meter (Figure 7c). The average density of barnacles on polymer PJZ9 plates was the smallest, while the average density of barnacles on polymer PD4 plates was the largest.

Taken together, the prepared polymers have excellent antifouling properties. The polymer PJ(D)0 has a significant inhibitory effect on mussels; therefore, it is necessary for us to

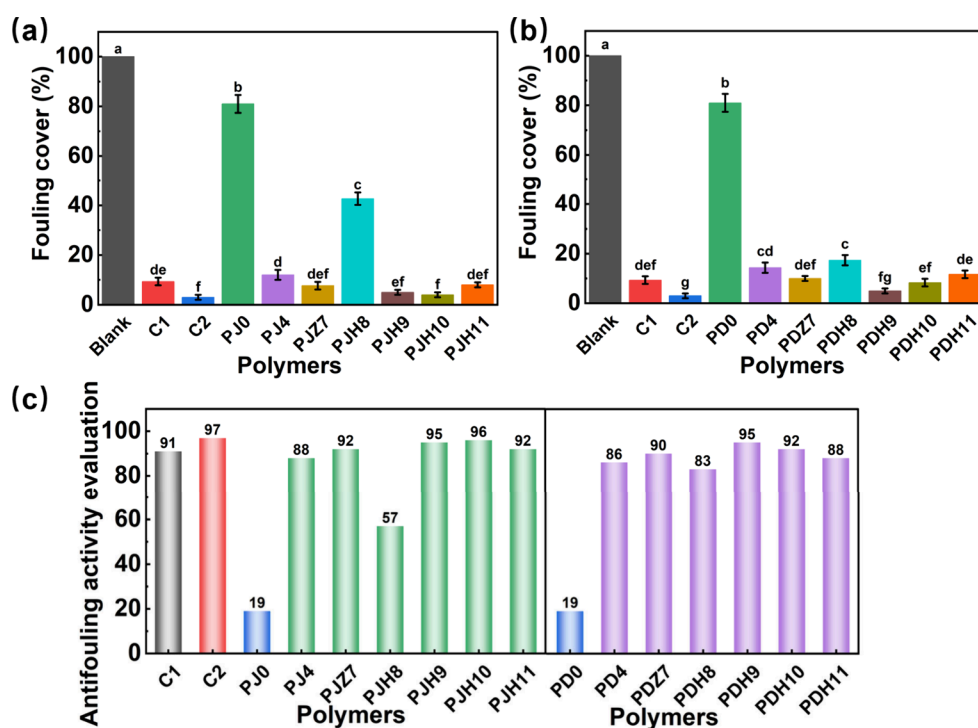


Figure 8. Relative percentage cover of fouling of polymer PJ (a) polymer PD (b) and on marine field test after 90 days. Bars, mean $\pm$ SE, and different lowercase letters indicate significant differences between groups ($P < 0.05$). (c) Evaluation of antifouling activity of polymers PJ and PD.

use acrylate as the matrix. The introduction of zinc monomers significantly inhibits the growth of algae, has a smooth surface, and is suitable for ship surfaces. The introduction of heterocyclic monomer HM enhances antifouling performance, but the surface is not smooth and is suitable for cage culture.⁵⁵

After 90 days of seawater immersion, the extent of biofouling on coated panels was quantified relative to the uncoated control (blank), as shown in Figure 8. Statistical analysis was conducted using OriginPro 2025b (Learning Edition). Data were first subjected to normality and homogeneity of variance tests. One-way ANOVA was performed to assess differences among groups, and Tukey's HSD test was used for pairwise comparisons. A significance threshold of $p < 0.05$ was applied. As illustrated in Figure 8a, the biofouling coverage on all experimental coatings (C1, C2, and PJ series polymers) was significantly lower than that of the uncoated control (blank), confirming strong antifouling activity ($p < 0.05$). The PJ-series polymer coatings notably reduced organism settlement compared to the blank, highlighting their effective antifouling performance. Similar trends were observed in Figure 8b for the PD series.

Figure 8c compares antifouling activity across all of the experimental and control groups. Polymers incorporating zinc-based or heterocyclic monomers exhibited significantly enhanced antifouling performance compared to standard formulations. Notably, PJH9 and PDH9, which contain heterocyclic monomer HM2, demonstrated antifouling efficacy comparable to the commercial copper oxide-based coating (C2). These results suggest that HM2-containing polymers hold strong potential as next-generation, environmentally friendly antifouling coatings.

3.7. Antifouling Mechanism of Polymer Materials. This study constructed a functional material system with excellent antifouling performance based on the PJ and PD series polymers. As shown in Figure 9, the antifouling mechanism of three types of polymer materials with different structural characteristics have significant differences.

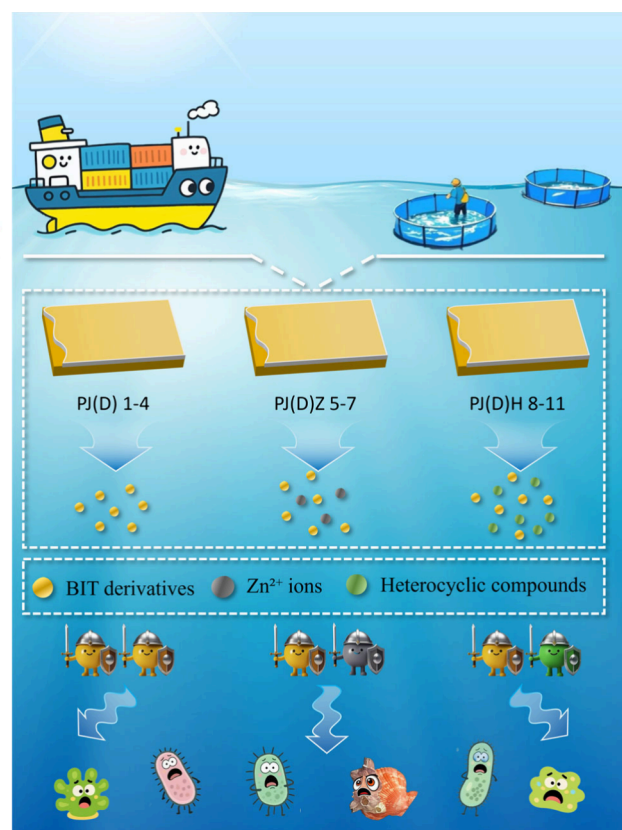


Figure 9. Schematic diagram of the antifouling mechanism of the polymer coating.

Specifically, polymers PJ(D)1–4 adopt a single BIT derivative active component system to achieve an antifouling function through the orientation of specific functional groups. It is noteworthy that polymers

PJ(D)Z5–7 and PJ(D)H8–11 both adopt a dual active component system: the former imparts the self-polishing characteristics of the material through the introduction of zinc-based monomers, while the latter achieves a synergistic double antifouling effect by introducing heterocyclic active groups.

4. CONCLUSIONS

Two BIT-triazine monomers were successfully synthesized by using cyanuric chloride, BIT and methylallyl alcohol (3-butenol) as starting materials. These monomers were copolymerized with methyl methacrylate (MMA), butyl acrylate (BA), and functional zinc or heterocyclic monomers to obtain a series of 22 BIT-triazine-based copolymers. Structural characterization and performance evaluations confirmed that the incorporation of zinc-containing or heterocyclic monomers significantly enhanced the antifouling efficacy of the coatings. The resulting polymers demonstrated excellent inhibitory effects against marine algae and bacteria, with inhibition rates reaching up to 99.93 and 99.69%, respectively. Remarkably, some coatings achieved 100% inhibition of barnacle larval attachment and mussel byssus formation. The polymer coatings exhibited strong environmental compatibility and long-term stability, making them promising candidates for a wide range of marine antifouling applications. Comparative analysis revealed that polymers derived from methylallyl alcohol (PJ series) showed a slightly superior performance compared to those synthesized with 3-butenol (PD series). This comprehensive study establishes clear structure–property relationships and demonstrates a modular synthetic strategy that enables precise tailoring of surface bioinertness via rational monomer selection. These findings offer a valuable design framework for the development of durable, eco-friendly antifouling coatings for marine environments.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsapm.5c01800>.

Additional experimental details, materials, and methods; ^1H NMR data and spectra; FTIR data and spectra; monomer composition ratio; molecular weight (M_n); PDI; T_g data; determination methods; linear regression equations; curves for three types of algal liquid standard curves (PDF)

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

AIBN, 2,2'-azobis(2-methylpropionitrile); BA, butyl acrylate; BIT, 1,2-benzisothiazoline-3-one; C1, 5-octyl-2-furanone; C2, cuprous oxide; *C. curvisetus*, *Chaetoceros curvisetus*; DSC, differential scanning calorimeter; FT-IR, Fourier transform infrared spectrometer; HM, heterocyclic monomers; HM1, benzo[d]-oxazol-2-yl acrylate; HM2, pyridin-2-yl acrylate; HM3, N-(6-chlorobenzo[d]thiazol-2-yl) acrylamide; HM4, N-(6-methylbenzo[d]thiazol-2-yl) acrylamide; *I. galbana*, *Isochrysis galbana*; MAA, methacrylic acid; M_n , number-average molecular weight; MMA, methyl methacrylate; NMR, nuclear magnetic resonance spectroscopy; PDI, polymer dispersion index; *S. aureus*, *Staphylococcus aureus*; TBT, tributyltin; T_g , glass transition temperature; TGA, thermal gravimetric analysis; TLC, thin layer chromatography; UV, ultraviolet; *V. corallilyticus*, *Vibrio corallilyticus*; *V. parahemolyticus*, *Vibrio parahemolyticus*; ZM, bifunctional zinc monomer

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